

v3 / v4 Integration Guide

Generated: 2026-05-18
Scope: How to connect the Material Properties Dashboard (v3) and Leak Rate Dashboard (v4)

1 — Repository Summary

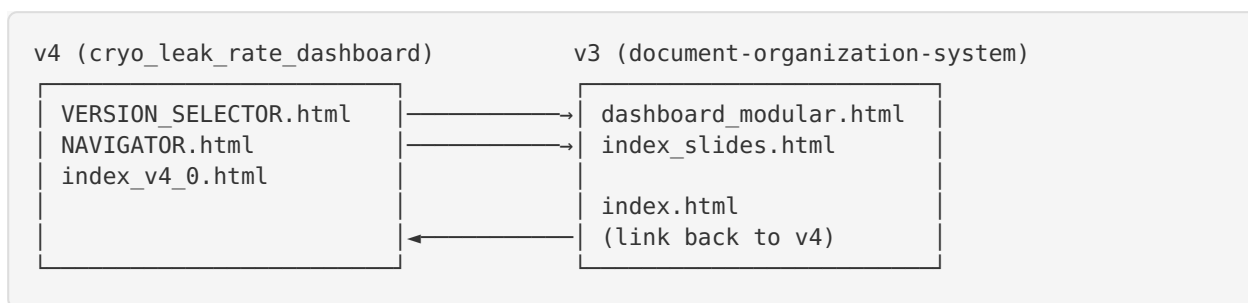
Property	v3 — Material Properties	v4 — Leak Rate Analysis
Repo	document-organization-system	cryo_leak_rate_dashboard
Version	v0.4.9	v4.0.0
GitHub	GBOGEB/document-organization-system (https://github.com/GBOGEB/document-organization-system)	⚠ Not yet created
Pages URL	gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/	gbogeb.github.io/cryo_leak_rate_dashboard/(planned)
Engine	JavaScript (browser-side)	Python (build-time)
Focus	k(T), cp(T), Y(T) for 10 materials	He leak → mass flow → cost
PRs	23 merged	Local only
Tests	823 NIST parity assertions	18 pytest tests
HTML files	7 interactive + 8 analysis	84 static

2 — Integration Strategy

Approach: Cross-Link (NOT Submodule)

- Rationale:**
- v3 and v4 are **different engineering domains** (material properties vs. leak rates)
 - v3 is **actively developed** (23 PRs, NIST parity tests, v0.4.9)
 - v3 updates independently and frequently
 - Submodule complexity is not justified

Integration method: Hyperlinks in both directions



3 — Step-by-Step Integration

Step 1: Verify v3 is clean and deployed

```
cd /home/ubuntu/github_repos/document-organization-system
git status          # Should be clean
git log --oneline -3
# Latest: 6f08258 – Merge PR #23

# Verify GitHub Pages is live:
# https://gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/
cryo_dashboard_v0_3_0/
```

Status: ☒ v3 is clean, pushed, and GitHub Pages is active.

Step 2: Create `cryo_leak_rate_dashboard` GitHub repo

The user must create this repo on GitHub:

1. Go to github.com/new (<https://github.com/new>)
2. Name: `cryo_leak_rate_dashboard`
3. Description: "MYRRHA QPLANT Cryogenic Leak Rate Analysis Dashboard v4.0.0"
4. Visibility: Private (or Public for GitHub Pages free tier)
5. Do NOT initialize with README (we have one)

Step 3: Push v4 to GitHub

```
cd /home/ubuntu/cryo_leak_rate_dashboard
git remote add origin https://github.com/GBOGEB/cryo_leak_rate_dashboard.git
git push -u origin main
```

Step 4: Deploy VERSION_SELECTOR.html

The `docs/VERSION_SELECTOR.html` is already created. To make it the root landing page:

Option A: Keep current structure (simpler)

```
docs/index.html → existing landing page (keep as-is)
docs/VERSION_SELECTOR.html → accessible at /VERSION_SELECTOR.html
```

Option B: Replace root index (requires build system update)

```
# Only do this after updating build scripts
cp docs/VERSION_SELECTOR.html docs/index.html
```

Step 5: Enable GitHub Pages for v4

```
Repository Settings → Pages
Source: Deploy from branch
Branch: main
Folder: /docs
```

Step 6: Add cross-link in v3

Optionally add a link from v3 back to v4. In `document-organization-system`, add to the v3 `index.html`:

```
<a href="https://gbogeb.github.io/cryo_leak_rate_dashboard/">
  → QPLANT Leak Rate Dashboard (v4.0.0)
</a>
```

4 — URL Map After Deployment

v4 URLs (cryo_leak_rate_dashboard)

URL	Content
<code>gbogeb.github.io/cryo_leak_rate_dashboard/</code>	Landing (index.html)
<code>.../VERSION_SELECTOR.html</code>	Version selector with v3/v4 cards
<code>.../NAVIGATOR.html</code>	Card-based entry point
<code>.../index_v4_0.html</code>	42-slide technical navigator
<code>.../STAKEHOLDER_PRESENTATION.html</code>	10-slide executive deck
<code>.../dashboard.html</code>	Triage dashboard
<code>.../calculations.html</code>	Detailed calculations
<code>.../rtm_traceability.html</code>	RTM traceability
<code>.../visualizations/*.html</code>	27 Plotly charts
<code>.../visualizations_v3/*.html</code>	22 advanced analytics
<code>.../plots/*.html</code>	5 core plots
<code>.../heroes/*.html</code>	10 audience pages
<code>.../compressors/*.html</code>	HP redundancy + WCS protection
<code>.../liquid_he/*.html</code>	Liquid operations guide
<code>.../archive/*.html</code>	3 archived presentations

v3 URLs (document-organization-system) — unchanged

URL	Content
<code>gbogeb.github.io/document-organization-system/cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0/</code>	v3 Landing
<code>.../dashboard_modular.html</code>	Interactive material properties
<code>.../material_properties_dashboard_v1_10.html</code>	Legacy fallback
<code>.../files.html</code>	File browser
<code>.../index_slides.html</code>	Presentation
<code>.../ssot_launcher.html</code>	SSOT launcher

5 — Testing Checklist

After deployment, verify all links work:

- ☐ `gbogeb.github.io/cryo_leak_rate_dashboard/` loads
- ☐ `VERSION_SELECTOR.html` displays both cards
- ☐ `v4 NAVIGATOR.html` card links work
- ☐ `v4 index_v4_0.html` loads with 42 slides
- ☐ `v4 STAKEHOLDER_PRESENTATION.html` loads with 10 slides
- ☐ `v4 visualizations` load (at least spot-check 3)
- ☐ `v4 hero pages` load (at least spot-check 2)
- ☐ `v3` link from `VERSION_SELECTOR` opens material properties dashboard
- ☐ `v3 dashboard_modular.html` is fully interactive
- ☐ `v3 → v4` cross-link works (if added)
- ☐ All relative paths in `v4 HTML` resolve correctly

6 — Future Considerations

Material Properties Integration into v4

If v4 ever needs to reference v3 material property data directly:

1. v3 exports CSV/JSON from its interactive dashboard
2. v4 can import those as `data/material_properties_export.json`
3. No need for submodule — just data file exchange

Shared SSoT Patterns

Both v3 and v4 follow the SSoT pattern:

- v3: `ssot.json` (material catalog, architecture, methods)
- v4: `data/config.yaml` + `data/*.json` (leak classes, scenarios, valves)

Version Evolution

- v3 may evolve to v0.5.x, v0.6.x (new materials, methods)
- v4 may evolve to v4.1.x (new analysis, updated specs)
- Both maintain independent version lineages